

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.metrics import confusion_matrix, classification_report

# 1. Load the dataset
# Ensure 'Wisconsin Breast Cancer dataset.csv' is in your folder
df_original = pd.read_csv('Wisconsin Breast Cancer dataset.csv')

# 2. Preprocessing
# We drop 'id' and 'diagnosis'
# We also drop any column that is completely empty (fixes the RuntimeWarning)
X = df_original.drop(['id', 'diagnosis'], axis=1)
X = X.dropna(axis=1, how='all')

# Convert 'M'/'B' to 1/0 for comparison
y = df_original['diagnosis'].map({'M': 1, 'B': 0})

# 3. Scaling (StandardScaler is mandatory for K-Means)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 4. Apply K-Means
kmeans = KMeans(n_clusters=2, random_state=42, n_init=10)
y_kmeans = kmeans.fit_predict(X_scaled)

# 5. Print Evaluation Metrics
```

```
print("Confusion Matrix:")
print(confusion_matrix(y, y_kmeans))
print("\nClassification Report:")
print(classification_report(y, y_kmeans))
```

6. PCA for Visualization

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
df_plot = pd.DataFrame(X_pca, columns=['PC1', 'PC2'])
df_plot['Cluster'] = y_kmeans
```

7. Visualization: K-Means with Centroids

```
plt.figure(figsize=(10, 7))
sns.scatterplot(data=df_plot, x='PC1', y='PC2', hue='Cluster', palette='Set1', s=100, alpha=0.7)
```

Transform centroids to PCA space

```
centers = pca.transform(kmeans.cluster_centers_)
plt.scatter(centers[:, 0], centers[:, 1], s=300, c='black', marker='X', label='Centroids')
```

```
plt.title('Experiment 10: K-Means Clustering (Cleaned Data)')
```

```
plt.legend()
```

```
plt.show()
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_breast_cancer
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.metrics import confusion_matrix, classification_report

# Load dataset
data = load_breast_cancer()
X = data.data
y = data.target

# Standardize data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Apply K-Means
kmeans = KMeans(n_clusters=2, random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)

# Evaluation
print("Confusion Matrix:")
print(confusion_matrix(y, y_kmeans))

print("\nClassification Report:")
print(classification_report(y, y_kmeans))
```

```
# PCA for visualization

pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)


# Create DataFrame

df = pd.DataFrame(X_pca, columns=['PC1', 'PC2'])

df['Cluster'] = y_kmeans

df['True Label'] = y


# Plot Clusters

plt.figure(figsize=(8, 6))

sns.scatterplot(data=df, x='PC1', y='PC2', hue='Cluster',
                palette='Set1', s=100, edgecolor='black', alpha=0.7)

plt.title('K-Means Clustering of Breast Cancer Dataset')

plt.xlabel('Principal Component 1')

plt.ylabel('Principal Component 2')

plt.legend(title="Cluster")

plt.show()


# Plot True Labels

plt.figure(figsize=(8, 6))

sns.scatterplot(data=df, x='PC1', y='PC2', hue='True Label',
                palette='coolwarm', s=100, edgecolor='black', alpha=0.7)

plt.title('True Labels of Breast Cancer Dataset')

plt.xlabel('Principal Component 1')

plt.ylabel('Principal Component 2')

plt.legend(title="True Label")

plt.show()


# Plot with Centroids

plt.figure(figsize=(8, 6))
```

```
sns.scatterplot(data=df, x='PC1', y='PC2', hue='Cluster',  
                palette='Set1', s=100, edgecolor='black', alpha=0.7)
```

```
centers = pca.transform(kmeans.cluster_centers_)  
plt.scatter(centers[:, 0], centers[:, 1],  
            s=200, c='red', marker='X', label='Centroids')
```

```
plt.title('K-Means Clustering with Centroids')  
plt.xlabel('Principal Component 1')  
plt.ylabel('Principal Component 2')  
plt.legend(title="Cluster")  
plt.show()
```